

Assignment 3 Report

The Effects of Iran War Risk on Global Financial Markets in 2026: a replication of Rigobon and Sack (2003) with NLP-selected war-news days

FRE-GY 7871 A · NLP and the Investment Process · Fall 2026

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GitHub repo: <https://github.com/Yuri48/FinEng> (folder HW3)

Data as of: 20 September 2026 (market data 2 January to 17 September 2026, 179 US business days)

1. Summary

This report re-runs the heteroskedasticity-based estimator of Rigobon and Sack (2003), *The Effects of War Risk on U.S. Financial Markets*, on the 2026 Iran war. The paper's key idea is that the effect of an unobservable "war risk" factor on asset prices can be identified without measuring the factor itself: one only needs a set of days on which the *variance* of war news was elevated. Rigobon and Sack chose those 17 days by reading newspapers. Here the days are chosen by an NLP pipeline that scores 26,436 Iran-related headlines, the daily Wikipedia *Current events* pages, and the Wikipedia day-by-day timeline of the war, and the estimator is then applied to 29 US and global financial variables.

Main findings:

- **Identification works through oil, not through Treasury yields, in 2026.** In 2003 the two-year Treasury yield was the natural numeraire: its variance was six times higher on war-news days. In 2026 the variance of the two-year yield is *lower* on the NLP-selected war days than on neighbouring days (0.0019 vs 0.0033), so the paper's normalisation is not identified over the full sample. The variance of the Brent front-month future, by contrast, triples on war days (19.5 vs 6.4 $\2). The 2026 results are therefore reported for a war-risk increase large enough to raise Brent by \$5 a barrel, with the paper's 25 bp-of-two-year-yield normalisation kept for the pre-war window, where it does hold.
- **Per \$5 of war-driven Brent (baseline, 45 war days):** WTI +\$5.1; S&P 500 -0.60%; Euro Stoxx 50 -1.2%; EM equities -1.3%; high-yield OAS +3.6 bp; BBB OAS +0.6 bp (marginal); 10-year yield +2.9 bp; 10-year break-even inflation +1.6 bp; broad dollar +0.20%; VIX +0.8 points; long Treasury ETF -0.29%; US airlines -2.5%; US energy stocks +1.5%; Henry Hub gas +1.4%. Gold, the yen, the Swiss franc, Bitcoin, the Nikkei, Tel Aviv 35 and the Saudi ETF show no significant response. All of the significant coefficients survive a tighter day cut-off (top 15%), the war/post-war sub-sample, and a hand-curated event list in the spirit of the paper's Table 1.
- **Compared with Iraq 2003, the sign of the yield and dollar responses flips.** In 2003 war risk pushed Treasury yields and break-even inflation *down* and the dollar *down* (flight into Treasuries, growth fears). In 2026 war risk pushed nominal yields and break-

evens *up* and the dollar *up*: the Strait of Hormuz closure made the conflict an inflationary oil-supply shock, and the dollar behaved as a safe-haven/energy-exporter currency. Credit spreads widen and equities fall in both episodes, and gold is unresponsive in both.

- **Equities were far less sensitive per unit of oil than in 2003.** A 2003 war shock that raised oil by 2% cut the S&P 500 by 3.8%; a 2026 war shock that raises Brent by about 5% cuts it by 0.6%. The 2026 shock was priced mainly as a commodity and inflation shock rather than as a recession risk; the S&P 500 rose about 10% over the sample despite the war.
- **The war-risk factor explains a large share of variance on war days:** at least 66% for WTI, 51% for Euro Stoxx 50, 48% for airlines, 42% for energy stocks, 35% for the dollar, 30% for high-yield spreads, 26-27% for the S&P 500, the 10-year yield and break-even inflation. Over the whole 8.5-month sample the shares are smaller (3-14%), because war-news days are one quarter of the sample and many other shocks were at work.
- **Pre-war window (2 January to 27 February, 14 of 39 days):** the closest analogue to the 2003 setting. There the two-year yield variance does rise on war days and the paper's normalisation gives a high-yield spread response of +37 bp and a BBB response of +14 bp per 25 bp fall in the two-year yield, close to the paper's +34 bp and +5 bp; the 10-year yield, break-even, equity and oil responses have the 2003 signs but are not statistically significant with so few days.

2. Method

2.1 The estimator (Rigobon and Sack 2003, equations 1-12)

Daily changes in a pair of financial variables are written as a reduced form driven by common factors and idiosyncratic shocks, $dx = D z + \eta$. The first factor z_1 is war risk; its loading on the first variable is normalised to one and its loading d_{21} on the second variable is the object of interest. Nothing about z_1 is measured. Instead, one splits the sample into "war news" days H, on which the variance of z_1 is high, and nearby "other" days L. If only the variance of z_1 differs between the two sets, and z_1 is orthogonal to the other factors, then the change in the covariance matrix is $d\Omega = \Omega_H - \Omega_L = d\text{Var}(z_1) \begin{bmatrix} 1, & d_{21}; & d_{21}, & d_{21}^2 \end{bmatrix}$, which gives two moment estimators, $d_{21} = d\text{Cov}/d\text{Var}(x_1)$ (eq. 7) and $d_{21} = d\text{Var}(x_2)/d\text{Cov}$ (eq. 6). Rigobon and Sack show these are IV estimators with instruments $\nu_1 = (+dx_1 \text{ on } H, -dx_1 \text{ on } L)$ and $\nu_2 = (+dx_2 \text{ on } H, -dx_2 \text{ on } L)$; combining the two instruments gives the third estimator, which the paper prefers. `warrisk.py` implements all three with heteroskedasticity-robust standard errors, plus the variance decomposition of the paper's Table 3: variance on L and H days, the predicted change in variance $d_{21}^2 d\text{Var}(x_1)$, the share of H-day variance explained (a lower bound on the war-risk share) and the share over the whole window under serial independence. Changes are demeaned within regime. L days are chosen as in the paper: the nearest business day to each H day that is not itself an H day, without reuse, so that the two sets have equal size and are as close in time as possible.

The identification assumption that matters most in 2026 is $d\text{Var}(x_1) > 0$: the variance of the numeraire variable must rise on war days. The tables report `dVar` for both numeraires and mark the variance decomposition as not identified when it fails.

2.2 Data

Financial variables (daily, 2 January to 17 September 2026): from FRED, the 2- and 10-year constant-maturity Treasury yields, 10-year break-even inflation, 10-year TIPS real yield, ICE BofA BBB and high-yield option-adjusted spreads, the Fed's broad trade-weighted dollar index and the VIX; from Yahoo Finance, S&P 500, Euro Stoxx 50, Nikkei 225, EEM (emerging markets), TA-35 (Israel), KSA (Saudi Arabia), ITA (US defense), JETS (US airlines), XLE (US energy), TLT (long Treasuries), HYG (high-yield bonds), BDRY (dry-bulk shipping), WTI and Brent front-month futures, gold and Henry Hub futures, DXY, EUR/USD, USD/JPY, USD/CHF and Bitcoin. Units follow the paper: percentage-point changes for yields and spreads, log-percentage changes for equities, currencies and ETFs, dollar changes for oil and gold. Changes are close-to-close on each series' own trading days and aligned to US business days (`market_data.py`).

News corpora (calendar days 1 January to 18 September 2026, `collect_news.py`, `wiki_timeline.py`): (i) Google News RSS headlines for two daily queries, a broad "Iran" query and a conflict query ("Iran war OR strikes OR missiles OR ceasefire OR Hormuz OR talks OR nuclear"), 26,436 unique headlines; (ii) the Wikipedia *Portal: Current events* page for every day, of which the Iran-related bullets are kept; (iii) the Wikipedia *Timeline of the 2026 Iran war*, split into its daily sections, extended back into January and February with dated sentences from the *Prelude*, *Negotiations* and *Oil market chronology* pages. GDELT's document API was rate-limited throughout and could not be used; the code handles it if it becomes available.

2.3 NLP pipeline (`nlp_pipeline.py`)

1. **Relevance filter.** A headline is relevant if it matches an Iran entity pattern (Iran, Tehran, IRGC, Khamenei, Hormuz, Houthi, Gulf states, Islamabad talks, ...) and a conflict/diplomacy pattern (war, strike, missile, blockade, ceasefire, talks, tanker, nuclear, sanctions, ...). 19,113 of the 26,436 headlines pass.
2. **Escalation lexicon.** Each relevant headline gets a score in $[-1, 1]$ equal to (escalation terms - de-escalation terms)/(total), with 70 escalation terms (strike, missile, blockade, ultimatum, mines, sinks, expires, ...) and 50 de-escalation terms (ceasefire, deal, talks, memorandum, reopen, stand-down, corridor, ...). The daily mean gives the *direction* of the news; the absolute day-to-day change in the mean is a *surprise-in-direction* feature.
3. **FinBERT tone.** Every relevant headline is scored with ProsusAI/FinBERT (a BERT model fine-tuned for financial sentiment); tone = $P(\text{negative}) - P(\text{positive})$. The daily mean is a second direction measure and the cross-headline standard deviation measures *disagreement* in coverage, which is a natural proxy for the variance of news.
4. **Novelty.** Headlines of each day are pooled into one document, TF-IDF vectorised (unigrams and bigrams, stop-words removed), and novelty is one minus the cosine similarity to the centroid of the previous five days. Days that introduce new vocabulary (a new theatre, a new instrument such as a "toll" or a "blockade", a new actor) are days with new information.
5. **Volume.** Number of relevant headlines, the share of Iran coverage that is conflict-related, and the character counts of the Iran-related Wikipedia current-events bullets and of the war-timeline section for that day.
6. **Composite index and mapping to trading days.** The seven z-scored features (volume, share, current-events volume, timeline volume, novelty, escalation swing, tone dispersion) are averaged into a daily index. Calendar days are mapped to the next US trading day (weekend and holiday news is attributed to Monday). H days are the top quantile of the index; L days follow the paper's nearest-day rule.

Because the RSS feed returns at most 100 items per query, the headline count saturates during the sustained-combat weeks of March; on those weeks the composite index is driven by the novelty, direction-swing and dispersion features, which are lower when every day brings the same kind of news. The index therefore favours days on which the *state* of the conflict changed (strikes begin, ceasefire, blockade, deal, collapse) over days of steady bombardment. A volume-only index (specification S7) is reported to show what happens when the sustained-war weeks are selected instead.

2.4 Specifications

Spec	Window	H days	Selection
S1 baseline	2 Jan - 17 Sep	45 of 179	top 25% of the composite NLP index
S2	same	27	top 15%
S3	same	45	top 25% of the index relative to its trailing 10-day median (surprise)
S4 pre-war	2 Jan - 27 Feb	14 of 39	top 36% (the paper's 17 of 47)
S5 war/post-war	2 Mar - 16 Sep	35 of 139	top 25%
S6 curated	2 Jan - 17 Sep	49	hand-collected event dates from the war timelines (the paper's method)
S7 volume	same	45	top 25% of a volume-only index

Each specification is estimated twice: normalised to a war-risk increase that lowers the two-year yield by 25 bp (the paper), and to one that raises the Brent front-month future by \$5.

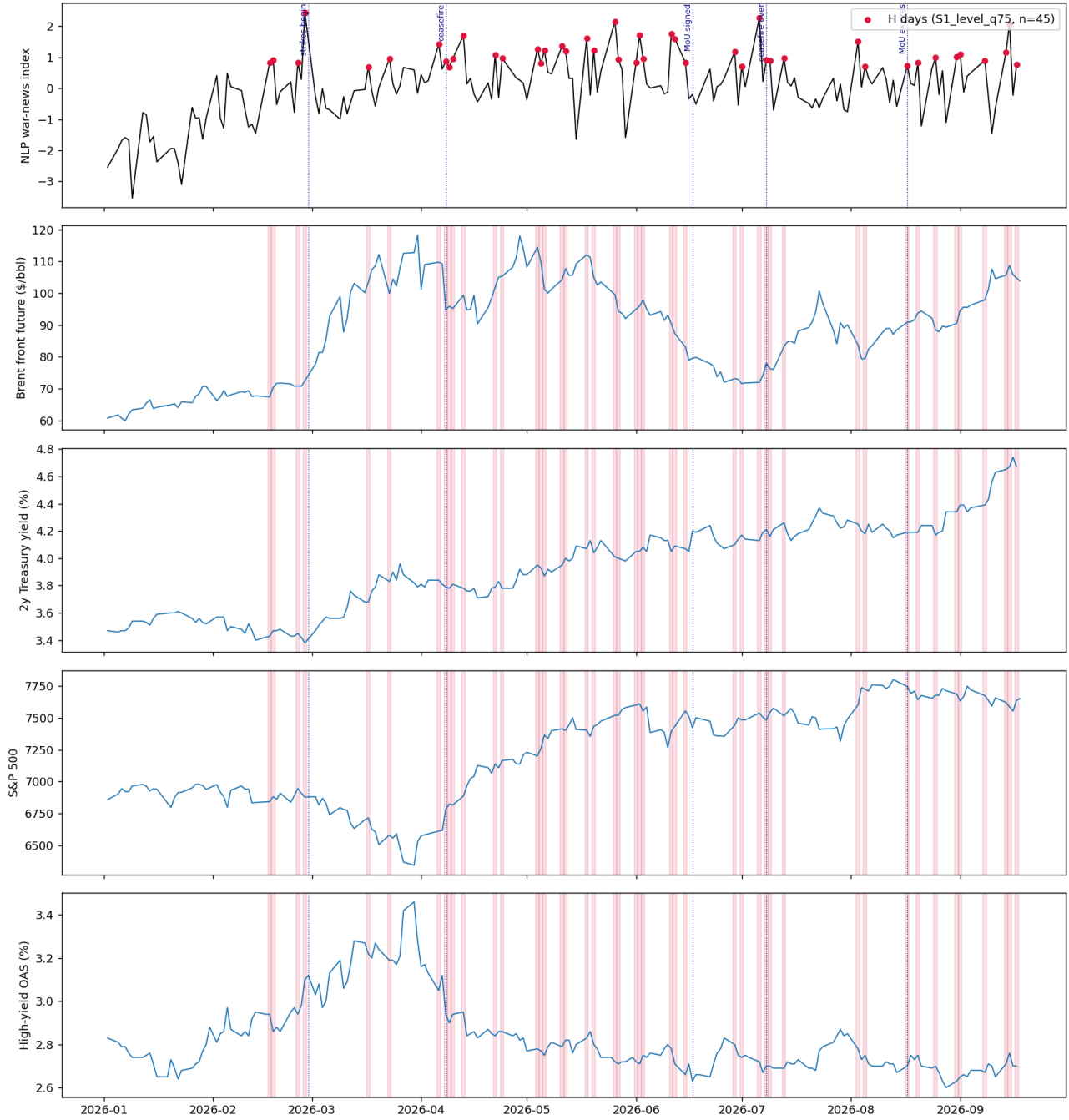
3. Results

3.1 The NLP-selected war-news days

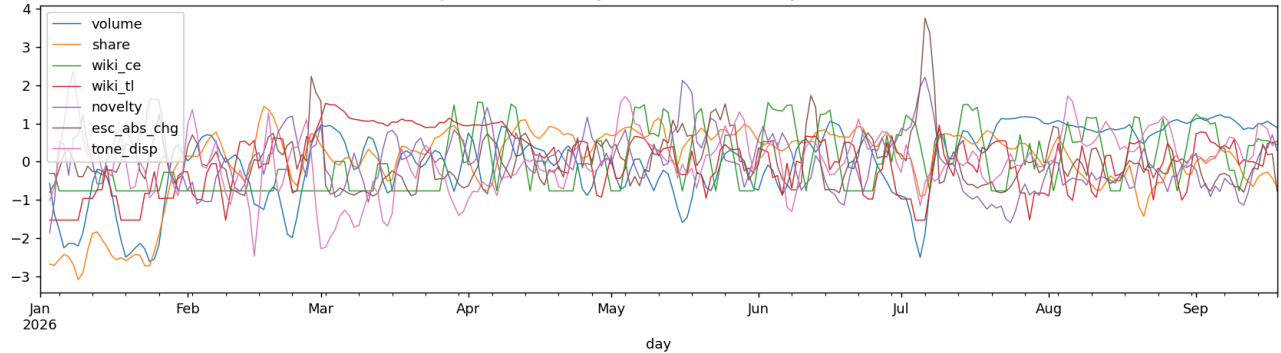
Figure 1 shows the composite index with the 45 baseline H days, against Brent, the two-year yield, the S&P 500 and the high-yield spread. The index is low and rising through January (protest crackdown, US naval build-up, on-off negotiations), spikes at the start of the strikes (27 February to 2 March), and then peaks at every change of state: the 8 April ceasefire and its collapse, the 13 April blockade, the May "deal or bombing" deadlines, the June Tomahawk exchanges and the 12-17 June memorandum, the 6-8 July tanker attacks that ended the ceasefire, the 17 August expiry of the memorandum, and the September tanker war. Table 1 lists each H day with its top headlines, the lexicon direction and the market moves of the day. The lexicon classifies 31 of the 45 days as risk-increasing, 12 as unclear and 2 as risk-decreasing; the lexicon is tilted toward escalation vocabulary, so several de-escalation days (12 and 15 June, 8 April) come out as "unclear" or "increased" because the headlines describe both the deal and the strikes that preceded it. The two clearly risk-decreasing days show the expected pattern: the two-year yield up, the S&P 500 up 1.1%, Brent down \$3.6, high-yield spreads 6 bp tighter.

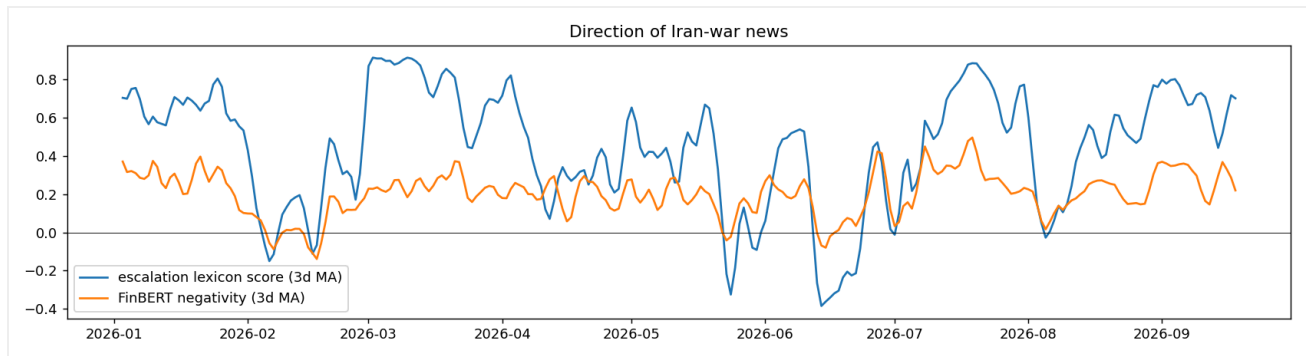
Seventeen of the 45 NLP days coincide with the 49 hand-curated event dates (12 would coincide by chance). The NLP set produces a sharper variance contrast than the curated set: the Brent variance ratio between H and L days is 3.1 for the NLP days and 2.2 for the curated days.

Iran war news intensity (NLP index) and markets, 2026 - shaded = high-variance (H) days



Components of the daily war-news index (3-day MA, z-scores)





3.2 Which variable carries the war-risk factor?

The paper's Table 3 logic can be turned around to ask which variables become more volatile on war days. Variance ratios (H over L, baseline days): Brent 3.1, Euro Stoxx 50 2.7, airlines 2.3, energy stocks 2.3, high-yield OAS 1.5, break-even inflation 1.3, dollar 1.3, gold 1.3, S&P 500 1.1, two-year yield 0.57, ten-year yield 0.63, TIPS real yield 0.41, VIX 0.76. Treasury yields were *less* volatile on war days. Two forces cancel: war news raises inflation expectations (break-evens co-move with Brent with correlation +0.69 on H days) and at the same time triggers flight to safety; in 2003 only the second force was present. The correlations of daily changes with Brent are all much stronger on H days than on L days (S&P 500 -0.68 vs -0.34, dollar +0.69 vs +0.29, high-yield +0.55 vs +0.20, two-year yield +0.58 vs +0.36), which is exactly the pattern the identification exploits, with Brent as the variable that loads on war risk.

Consequently the two-year-yield normalisation is not identified over the full 2026 sample (negative $dVar$, unstable and insignificant coefficients in every full-sample specification; see the S1, S5, S6 tables), while the Brent normalisation is well identified and stable across specifications.

3.3 Sensitivities to war risk (Brent normalisation)

Table 2b reports the baseline. Per \$5 of war-driven Brent (combined-instrument IV, $|t|$ in parentheses): WTI +\$5.14 (8.2); 10-year yield +2.9 bp (3.0); break-even inflation +1.6 bp (3.7); TIPS real yield +1.4 bp (1.4); S&P 500 -0.60% (3.7); Euro Stoxx 50 -1.20% (4.5); EEM -1.29% (3.4); Nikkei -0.6% (0.7); high-yield OAS +3.6 bp (2.7); BBB OAS +0.6 bp (1.7); HYG -0.19% (2.7); TLT -0.29% (2.2); broad dollar +0.20% (2.8); DXY +0.16% (2.7); EUR/USD -0.16% (1.2); USD/JPY and USD/CHF nil; gold -\$4 (0.2); VIX +0.82 (2.1); airlines -2.5% (4.7); energy +1.5% (3.1); defense ETF -1.1% (2.7); natural gas +1.4% (2.3); shipping +0.7% (1.1); TA-35 +0.4% (1.0); KSA -0.45% (1.5); Bitcoin -0.4% (0.6).

Three of these deserve comment. First, the *defense* ETF falls with war risk, the opposite of the popular narrative; in 2026 defense stocks traded as high-multiple growth stocks and fell with the broad market, and the days with the most escalatory news were also days of talk about the fiscal cost of the war. Second, Europe and emerging markets are more sensitive than the US (Euro Stoxx -1.2% and EEM -1.3% against -0.6% for the S&P 500), consistent with their greater dependence on Gulf energy. Third, the dollar strengthens, which together with the rise in yields distinguishes 2026 from 2003: the US was a net energy exporter in 2026, and Treasury yields rose rather than fell.

The results are robust. With the top 15% of days (S2) the significant coefficients keep their signs and magnitudes (S&P 500 -0.47%, Euro Stoxx -0.82%, 10-year +3.5 bp, dollar +0.14%). Over the war

and post-war window (S5) t-statistics rise (S&P 500 -0.61% with $t = 4.6$, Euro Stoxx -1.18% with $t = 4.9$, 10-year +3.0 bp with $t = 3.7$, break-even +1.7 bp with $t = 4.5$, high-yield +3.3 bp with $t = 2.9$, dollar +0.22% with $t = 3.5$). The hand-curated days (S6) give S&P 500 -0.70%, high-yield +4.6 bp, 10-year +2.6 bp, break-even +1.7 bp, Euro Stoxx -0.93%, airlines -1.9%, energy +1.0%, all significant, and a dollar response of +0.11% that is not. The surprise index (S3) yields weaker identification (Brent variance ratio 2.0) and only the oil-linked responses (WTI, airlines) remain significant: it is the *level* of war-news intensity, not its change relative to the recent past, that marks the high-variance days. The volume-only index (S7), which selects the sustained-combat weeks of March, gives a Brent variance ratio of only 1.8 and noisier estimates with the same signs (Euro Stoxx 50 -1.25% with $t = 3.2$, energy stocks +1.9% with $t = 3.0$, high-yield +5.1 bp with $t = 2.0$, S&P 500 -0.47% with $t = 1.4$): when every day is a war day, the nearest "other" days are war days too and the variance contrast that the estimator needs is diluted.

3.4 Variance decomposition

Table 3 for each specification reproduces the paper's Table 3. On baseline war days the war-risk factor accounts for at least 66% of the variance of WTI, 51% of Euro Stoxx 50, 48% of airlines, 42% of energy stocks, 35% of the broad dollar, 30% of high-yield spreads, 27% of the 10-year yield and of break-even inflation, 26% of the S&P 500, 23% of EEM and 22% of the VIX and the defense ETF, and essentially none of gold, the yen, the franc or Bitcoin. Over the full sample (179 days, 45 of them war days) the corresponding lower bounds are 14% for WTI, 9% for Euro Stoxx 50 and airlines, 7% for energy stocks, 3-4% for the S&P 500, the dollar, the 10-year yield, break-evens and high-yield spreads. The paper found 13-63% over ten weeks in which 17 of 47 days were war days; the 2026 sample is 3.8 times longer and contains other large shocks, so the smaller full-sample shares are expected.

3.5 The pre-war window and the two-year-yield normalisation

Between 2 January and 27 February the conflict was a *risk*, not a war, which is the paper's setting. The two-year yield variance rises on the 14 NLP-selected days (0.0017 vs 0.0011), so the paper's normalisation is identified. For a war-risk increase that lowers the two-year yield by 25 bp: 10-year yield -8 bp (0.4), break-even -10 bp (1.4), TIPS real yield -15 bp (1.5), S&P 500 -0.5% (0.1), BBB OAS +14 bp (2.1), high-yield OAS +37 bp (2.5), WTI -\$0.4 (0.05), gold not identified, broad dollar -1.4% (0.7), Tel Aviv 35 -5.1% (1.3), airlines -15% (1.9), Bitcoin -33% (2.7). The credit-spread responses are strikingly close to 2003 (+34 bp high-yield, +5 bp BBB), and every point estimate for yields, equities and the dollar has the 2003 sign, but with 14 days the standard errors are large and the oil response, the hallmark of the later sample, is absent: in January and February oil moved with the news about talks (the -4% on 15 January, the -3% on 5 and 12 February, the +4% on 18 February) but its variance rose only by half on war days, so the pre-war identification runs through rates and credit rather than through oil.

3.6 Comparison with Iraq 2003

	Iraq 2003 (per -25 bp two-year yield)	Iran 2026 (per +\$5 Brent)	Sign
10-year Treasury yield	-26 bp (11.9)	+2.9 bp (3.0)	flips
Break-even inflation	-11 bp (3.5)	+1.6 bp (3.7)	flips
S&P 500	-3.76% (2.9)	-0.60% (3.7)	same
BBB spread	+5 bp (3.8)	+0.6 bp (1.7)	same
High-yield spread	+34 bp (5.4)	+3.6 bp (2.7)	same

	Iraq 2003 (per -25 bp two-year yield)	Iran 2026 (per +\$5 Brent)	Sign
Oil	+\$0.77 12-month WTI (2.4)	+\$5.14 front WTI (8.2)	same
Gold	+\$1.3 (0.3)	-\$4 (0.2)	none in both
Dollar (broad)	-0.44% (2.2)	+0.20% (2.8)	flips

To put the two episodes on a common footing, scale each shock by the oil move it induces. The 2003 shock moved 12-month WTI by about 2% of its price and cut the S&P 500 by 3.8%; the 2026 shock moves Brent by about 5% and cuts the S&P 500 by 0.6%. Per percent of oil, US equities were roughly ten times more sensitive to war risk in 2003. The 2003 war risk was priced as a growth shock with a modest energy component (Iraq's output was small and the Gulf stayed open); the 2026 war risk was priced as an energy-supply and inflation shock (the Strait of Hormuz was closed or contested for most of the sample) with a small growth component. That is also why the sign of the yield, break-even and dollar responses flips: in 2003 investors bought Treasuries and sold dollars on war news; in 2026 they sold Treasuries on inflation fears and bought dollars.

4. Limitations

- **Timing.** News is attributed to the next US trading day by calendar date. Events after the US close on a weekday (the first strikes began around 18:30 ET on Friday 27 February) are attributed to that day rather than to the next trading day; Rigobon (2003, section IV) shows that misplacing some high-variance days lowers the variance contrast but does not bias the estimator as long as the H set still has higher war-news variance on average.
- **Headline volume is censored** at 100 items per query per day by the RSS feed, which mutes the volume component during March; the volume-only specification S7 documents the effect.
- **Sustained-war contamination of L days.** During March almost every day carried war news, so "nearest other days" are themselves war days. This works against finding a variance contrast and is one reason the composite index, which rewards changes of state, identifies better than raw volume.
- **The escalation lexicon is hand-built** and tilted toward escalation terms; it is used only to label the direction of H days (Table 1), not in the estimator.
- **FinBERT** was trained on financial phrase-bank sentences, not on war headlines; its tone is used only as a disagreement feature and a direction check.
- **Other factors are assumed homoskedastic** across H and L days. Nearest-day matching mitigates this, but 2026 also contained large monetary-policy and fiscal news; days on which those coincided with war news add noise to the two-year-yield results in particular.
- **GDELT** could not be pulled (rate-limited), so the index has no measure of global coverage share; the code uses it automatically when available.

5. Files

README.md describes the pipeline and AI_USE.md discloses the use of AI. Scripts: market_data.py, collect_news.py, wiki_timeline.py, nlp_pipeline.py, warrisk.py, run_analysis.py, annotate_days.py, make_report.py, build_html.py. Data: data/. Outputs: output/table1_*.csv (H days with headlines), output/table2_*_ust2y.csv and output/table2_*_brent.csv (all estimates and

variance decompositions for every specification), `output/summary_across_specs_*.csv`, `output/H_days_*.csv`, `output/L_days_*.csv`, `output/index_trading_days.csv`, `output/fig/`.

References

Rigobon, R. (2003). Identification through heteroskedasticity. *Review of Economics and Statistics* 85(4), 777-792. Rigobon, R. and B. Sack (2003). The effects of war risk on U.S. financial markets. NBER Working Paper 9609. Araci, D. (2019). FinBERT: Financial sentiment analysis with pre-trained language models (ProsusAI/finbert).

Appendix: tables

Table 1. NLP-selected high-variance war-news days (baseline S1)

Direction = sign of the escalation-lexicon score of that day's relevant headlines (>0.2 increased risk, <-0.2 decreased). Market columns are the day's changes: 2y yield (bp), S&P 500 (%), Brent front future (\$), high-yield OAS (bp).

date	index	risk_direction	d2y_bp	dSPX_pct	dBrent_usd	dHY_bp	headlines		
2026-02-17	0.82	Unclear	3	0.10	-0.33	0	Iran says 'guiding principles' agreed with US at nuclear talks	Iran says "clearer path ahead" to nuclear deal with U.S. after talks in Geneva under shadow of Trump's threats	Reza Pahlavi urges Trump to ditch Iran talks as US e
2026-02-18	0.92	Increased	4	0.56	2.93	-8	President Trump warns of 'bad things' if Iran fails to make nuclear deal	Iran says 'good progress' made in nuclear talks with US in Geneva	Trump warns of 'bad things' if Iran doesn't make a deal, as U.S. carrier approaches reg
2026-02-25	0.83	Unclear	2	0.81	0.08	-3	U.S. and Iran wrap up 'most intense' nuclear talks with no deal — more negotiations ahead	Iran threatens escalation if US attacks	US-Iran nuclear talks end without a deal as threat of war grows
2026-02-27	2.44	Increased	-4	-0.43	1.73	12	Attacks on Iran and retaliatory strikes 'undermine international peace and security'	Iranian leader Khamenei killed in air strikes as U.S., Israel launch attacks	Mapping US and Israeli attacks on Iran and Tehran's retaliatory
2026-03-17	0.68	Increased	0	0.25	3.21	-5	Iran Launches Missile Attack on Tel Aviv in Retaliation for Latest Killings	Strikes hit world's largest natural gas field in Iran, and Tehran retaliates with more attacks	Iran blames Israel for gas field attack, fires missiles
2026-03-23	0.96	Increased	-5	1.14	-12.25	-5	Iran strikes near Israeli nuclear research center as Trump threatens attacks on Iranian power plants	Trump says Iran is eager for a deal to end the war as he extends deadline to allow for diplomacy	Iran attacks near Israeli nu
2026-04-06	1.43	Increased	0	0.44	0.74	-8	Trump warns of "critical period" in Iran war, threatening severe strikes if there's no deal by Tuesday night	Iran updates: Pakistan seeks 2-week pause after Trump warns 'whole civilization will die' if no deal by deadline	Trum

date	ind ex	risk _dir _ex _ction	d2 y_ bp	dS PX _pc _t	dBr ent _us _d	dH Y_ bp	headlines		
2026-04-08	0.88	Increased	-2	2.48	-14.52	-18	War on Iran during nuclear negotiations undermines the US's ability to talk peace around the world – and the effects won't end when Trump leaves office	Ceasefire is threatened as Israel expands Lebanon strikes and Iran closes st	
2026-04-09	0.69	Increased	-1	0.62	1.17	-4	War on Iran during nuclear negotiations undermines the US's ability to talk peace around the world – and the effects won't end when Trump leaves office	Iran war ceasefire teeters over disagreements on Lebanon and the Strait of H	
2026-04-10	0.96	Unclear	3	-0.11	-0.72	4	US and Iran end 21-hour ceasefire talks without agreement before Vance departs Pakistan	U.S. and Iran prepare for ceasefire talks as Netanyahu authorizes negotiations with Lebanon	How many people have been killed in the Iran w
2026-04-13	1.70	Unclear	-3	1.01	4.16	1	Trump threatens Strait of Hormuz blockade after US-Iran ceasefire talks end without agreement	US begins blockade of Iran's ports, Tehran threatens retaliation	US military says it will blockade Iranian ports after ceasefire tal
2026-04-22	1.07	Increased	1	1.04	3.43	-1	Iran says it has seized two ships in Strait of Hormuz after U.S. extends ceasefire	Iran fires on 3 ships in the Strait of Hormuz, complicating efforts to resume U.S. ceasefire talks	Trump extends ceasefire as uncertainty over
2026-04-24	0.97	Increased	-5	0.79	0.26	0	Latest ceasefire talks fail as Iran's top diplomat leaves Pakistan and Trump tells envoys not to go	Trump cancels US envoys' trip to Pakistan for talks on Iran war	Live Updates: Hopes for Peace Deal Rise After Iran Says Strait
2026-05-04	1.27	Increased	7	-0.41	6.27	1	U.S. And Iran Fail to Agree on Peace Deal After 21 Hours of Talks, Vance Says	Markets on edge as fresh U.S.-Iran attacks dent optimism over a peace deal	US strikes Iranian fast boats as Iran attacks UAE oil facility
2026-05-05	0.80	Increased	-2	0.81	-4.57	-1	Trump says Iran will be bombed at a 'much higher level' if it doesn't agree to peace deal	U.S. And Iran Fail to Agree on Peace Deal After 21 Hours of Talks, Vance Says	Markets on edge as fresh U.S.-Iran attacks dent optimism o
2026-05-06	1.21	Increased	-6	1.45	-8.60	-2	Trump says Iran will be bombed at a 'much higher level' if it doesn't agree to peace deal	Trump declares US has won the war with Iran as nuclear deal negotiations continue	Iran War Updates: Tehran and U.S. Offer Conflicting Me
2026-05-11	1.36	Increased	5	0.19	2.92	-2	Iran says US making 'unreasonable' demands in negotiations to end war	Iran warns the US against attacks on its oil tankers and other ships but ceasefire appears to hold	Israel is worried that Trump will strike a 'bad deal' wit
2026-05-12	1.19	Increased	5	-0.16	3.56	3	Iran threatens to "teach a lesson" if U.S. attacks, Trump says ceasefire is "on life support"	Israel is worried that Trump will strike a 'bad deal' with Iran, leaving war objectives unmet	U.S. Might Restart Strikes on Iran, Tr
2026-05-18	1.62	Increased	-2	-0.07	2.84	3	Trump calls off scheduled attack on Iran amid "serious negotiations" toward peace deal	Iran war day 78: Trump, Tehran signal talks as Lebanon truce extended	U.A.E. reports drone strike at nuclear power plant as Iran war deadlo
2026-05-20	1.21	Increased	-9	1.07	-6.26	-6	Iran Threatens to Strike Beyond the Middle East if the U.S. Resumes Attacks	Iran threatens to extend conflict 'beyond the region' if U.S. and Israel resume attacks	Iran threatens war 'beyond the region' if U.S. attacks
2026-05-26	2.16	Unclear	-12	0.61	-3.96	-2	U.S. military strikes Iran as Trump says negotiations move forward for deal to end war	U.S. and Iran work toward deal to extend ceasefire and reopen Strait of Hormuz	US says it launched 'self-defense strikes' in Iran as peace
2026-05-27	0.93	Unclear	-1	0.02	-5.29	-1	Traders' hopes fade for U.S.-Iran nuclear deal this year despite report on potential ceasefire agreement	The Latest: Iran negotiators agree to extend ceasefire, begin nuclear talks pending Trump approval	U.S. and Iran Move Tow

date	indirection	risk_direction	d2y_bp	dSPX_pct	dBrEnt_usd	dHY_bp	headlines		
2026-06-01	0.83	Unclear	7	0.26	2.93	-2	Iran fires missiles and US strikes Iran facility after reports of faltering peace talks	U.S. bombs Iranian military sites and downs missiles Tehran fired at troops in Kuwait	US Bombs Iranian Military Sites, Then Downs Missiles
2026-06-02	1.72	Increased	0	0.13	1.02	-1	Iran fires missiles and US strikes Iran facility after reports of faltering peace talks	Kuwait says Iranian drones hit airport and killed 1 as ceasefire is tested again	One killed and dozens injured in Iranian drone strikes on
2026-06-03	0.95	Increased	3	-0.74	1.81	4	Kuwait says Iranian drones hit airport and killed 1 as ceasefire is tested again	One killed and dozens injured in Iranian drone strikes on Kuwait airport	Iranian drone attack kills Indian citizen in Kuwait after US strikes Qes
2026-06-11	1.76	Unclear	-8	1.74	-2.72	-2	Iran war day 104: Iran attacks US bases, closes strait after Trump strikes	US and Iran have agreed to wording of a deal to end their war, Pakistan's prime minister says	Trump calls off latest threats to strike Iran, citing a b
2026-06-12	1.60	Decreased	4	0.50	-3.05	-7	Trump condemns Israeli strike in Beirut, warning attacks threaten deal on U.S-Iran war	US and Iran have agreed to wording of a deal to end their war, Pakistan's prime minister says	Trump calls off latest threats to strike Iran
2026-06-15	0.83	Decreased	-2	1.64	-4.16	-5	Iran and US agree deal to open Strait of Hormuz and extend ceasefire	U.S. and Iran Reach Agreement to Reopen Strait and Begin Nuclear Talks	Trump condemns Israeli strike in Beirut, warning attacks threaten deal on U.S-Iran war
2026-06-29	1.19	Unclear	3	1.17	1.16	-3	Iran attacks Bahrain and Kuwait following US strikes and threatens to halt talks to end the war	Iran attacks Bahrain and Kuwait following US strikes and threatens to halt talks	Iran attacks Bahrain and Kuwait following U.S. st
2026-07-01	0.71	Unclear	3	-0.22	-1.35	-1	Iran Bans UN Nuclear Inspectors from Bombed Nuclear Sites after US-Israel Strikes	U.S. Military Strikes Missile and Drone Sites in Iran	As the Pentagon stays quiet, AP reconstructs a US strike that killed over 100 Iranian chil
2026-07-06	2.28	Increased	-1	0.72	0.19	-2	U.S. Strikes Iran and Reimposes Sanctions in Retaliation for Tanker Attacks	US Iran strikes today: US launches new strikes on Iran, revokes oil sales permit after 3 ships attacked in Strait of Hormuz	US launches new strikes on
2026-07-08	0.91	Increased	2	-0.28	3.86	3	U.S. hits dozens of Iranian targets in retaliatory strikes after ship attacks in Strait of Hormuz	Iran fires 10 missiles at Jordan after US strikes reported near Bushehr nuclear plant	US Iran strikes today: US launches new str
2026-07-09	0.89	Increased	-5	0.81	-1.72	0	Iran fires 10 missiles at Jordan after US strikes reported near Bushehr nuclear plant	Iran Says U.S. Strikes Targeted Bushehr Nuclear Plant, Warns of Retaliation Against American Bases	US and Iran exchange intensifying fire ac
2026-07-13	0.97	Increased	5	-0.80	7.29	0	U.S.-Iran strikes escalate over the Strait of Hormuz, threatening to collapse ceasefire	Trump resumes Iran port blockade and threatens strikes on energy targets	Iran launches missiles and drones at Gulf states after US strikes
2026-08-03	1.50	Unclear	-3	1.47	-6.35	-7	Iran denies seeking halt to US attacks, agreement to reopen Strait of Hormuz	Trump threatens more strikes on Iran. Tensions from Hormuz to Kuwait and Gaza lead to more warnings	Iran threatens to strike other nations' energy fi
2026-08-05	0.71	Unclear	-2	-0.17	0.09	2	Trump warns Iran will be hit 'really hard' if nuclear negotiations collapse again	Trump warns Iran will be hit 'really hard' if nuclear negotiations collapse again	EXCLUSIVE: Iran threatens to hit Gulf states if US launches ne
2026-08-17	0.73	Increased	2	-0.52	2.35	3	As U.S.-Iran deadline for broad peace deal expires, Trump threatens to 'bomb' Oman	Iran threatens new offensive while US rules out extending ceasefire deal	Trump won't extend Iran ceasefire, threatens to 'bomb' Oman if it 'get

date	ind ex	risk _dir _ex _ction	d2 y_ bp	dS PX _pc _t	dBr ent _us _d	dH Y_ bp	headlines		
2026-08-20	0.82	Increased	0	-0.87	2.16	2	Iran vows 'devastating' response as US threatens crushing sanctions designed to 'collapse' regime	US allies in Asia wary as Trump moves military assets for Iran war	Iran threatens military response to US sanctions
2026-08-25	0.99	Increased	-7	0.32	-3.59	1	Iran's nuclear chief says attacked nuclear sites not secure for IAEA inspection	Can Trump's economic war against Iran do what airstrikes and negotiations couldn't?	Iran vows retaliation after U.S. widens sanctions, fueling fea
2026-08-31	1.01	Increased	0	-0.33	1.18	3	Iran claims attacks on Bahrain, Jordan, Iraq after US strikes kill 11	Iran fires on its Gulf neighbors, retaliating for US strikes after a wedding was hit	U.S. Attacks Island in Strait of Hormuz; Iran Retaliates With Missile F
2026-09-01	1.11	Increased	5	-0.71	4.16	2	Iran claims attacks on Bahrain, Jordan, Iraq after US strikes kill 11	Iran fires on U.S. allies in Gulf after night of American strikes it claims killed four at a wedding	Iran fires on its Gulf neighbors, retaliating for US st
2026-09-08	0.90	Increased	2	-0.59	1.64	-1	US military strikes three Iranian tankers in retaliation for missile attacks	Hormuz traffic slows after Iran threatens retaliation for US attacks	U.S. Hits 5 Iranian Oil Tankers, Citing Attempted Strikes on Warship
2026-09-14	1.16	Increased	2	-0.48	1.07	6	Iran-backed Houthis claim to hit Saudi base with missiles, drones amid renewed fighting	Iraq probes drone strikes on Saudi Arabia, shuts three crossings to Iran	Iran hardliners said to have launched attacks in defiance of lead
2026-09-15	2.07	Increased	2	-0.45	3.07	5	Saudi strikes and Houthi attacks widen Middle East war	New photos show widespread damage at U.S. positions caused by Iranian missile, drone attacks	Iran war cost hits \$38 billion, forecast to rise \$3 billion a month, CBO says
2026-09-17	0.77	Increased	-7	1.13	-1.01	0	Iran in, Abbas out: Trump threatens new strikes as US grants Tehran leaders UN visas	Houthi media says Saudi strikes kill one, hit telecom towers in Yemen	US approves visas for top Iranian leaders to attend UN high-level meeti

Table 2 (S1_level_q75). Estimated impact of an increase in war risk, normalised to a 25 bp drop in the two-year Treasury yield

S1 baseline: full sample, top-25% NLP index, nearest-day L. Window 2026-01-02 to 2026-09-17; nH = 45, nL = 45. Two-year yield variance: L days 0.00334, H days 0.00190. Absolute t-statistics (heteroskedasticity-robust) in parentheses.

Variable	Units	IV with nu1	IV with nu2	IV with nu3
Ten-year Treasury yield	pp chg	-0.143 (1.83)	-0.333 (2.17)	-0.0667 (0.88)
10y break-even inflation	pp chg	0.111 (0.87)	-0.0738 (0.88)	0.0177 (0.44)
S&P 500	pct chg	1.59 (0.84)	2.72 (0.37)	1.5 (0.87)
BBB corporate OAS	pp chg	0.0454 (1.40)	-0.0483 (0.55)	0.00957 (0.28)
High-yield corporate OAS	pp chg	0.219 (1.15)	-0.2 (0.73)	0.00993 (0.07)
WTI front-month future	\$ chg	15.3 (0.74)	-60.6 (1.67)	-17 (1.19)
Gold future	\$ chg	231 (1.14)	-303 (0.50)	117 (0.70)
Broad trade-weighted dollar	pct chg	-0.658 (1.32)	0.79 (0.25)	-0.616 (1.23)
10y TIPS real yield	pp chg	-0.254 (2.37)	-0.252 (3.84)	-0.252 (3.89)
Brent front-month future	\$ chg	16.2 (0.76)	-52.7 (2.09)	-15.3 (1.43)
DXY dollar index	pct chg	-1.53 (1.91)	-1.64 (1.88)	-1.59 (2.22)

Variable	Units	IV with nu1	IV with nu2	IV with nu3
EUR/USD	pct chg	-0.475 (0.64)	0.645 (0.09)	-0.445 (0.59)
USD/JPY	pct chg	-0.101 (0.11)	-91.1 (0.10)	-0.432 (0.42)
USD/CHF	pct chg	0.201 (0.23)	5.83 (0.15)	0.168 (0.19)
VIX	pt chg	-3.54 (0.84)	-9.88 (0.86)	-3.97 (0.87)
20y+ Treasury ETF (TLT)	pct chg	0.0299 (0.02)	258 (0.02)	-1.6 (0.87)
HY bond ETF (HYG)	pct chg	-0.118 (0.11)	22.1 (0.13)	1.11 (1.94)
Euro Stoxx 50	pct chg	-4.73 (0.86)	12.6 (1.63)	3.32 (0.89)
Nikkei 225	pct chg	-1.58 (0.51)	-23.3 (0.42)	-1.85 (0.57)
EM equities (EEM)	pct chg	0.897 (0.16)	-5.95 (0.04)	1.23 (0.38)
Tel Aviv 35	pct chg	1.15 (0.31)	-34 (0.25)	0.439 (0.11)
Saudi equities (KSA ETF)	pct chg	-1.01 (0.32)	18.2 (0.45)	0.703 (0.31)
US defense ETF (ITA)	pct chg	-3.11 (0.51)	16.1 (0.96)	0.899 (0.29)
US airlines ETF (JETS)	pct chg	-10.7 (0.76)	23.1 (2.32)	5.59 (1.29)
US energy ETF (XLE)	pct chg	12 (0.98)	-9 (2.44)	-2.2 (0.82)
Henry Hub natgas future	pct chg	10.6 (0.96)	-8.54 (0.54)	2.89 (0.46)
Dry-bulk shipping ETF (BDRY)	pct chg	6.48 (0.76)	-35 (1.02)	-3.64 (0.51)
Bitcoin	pct chg	12.5 (1.45)	-6.86 (0.51)	2.19 (0.31)

Table 3 (S1_level_q75). Variances and share explained by the war-risk factor

Variable	Var. on L days	Var. on H days	Predicted change in var.	% explained, H days	% explained, all days
Ten-year Treasury yield	0.002573	0.001624	-0.0001021		
10y break-even inflation	0.0003911	0.00052	-7.161e-06		
S&P 500	0.6766	0.7196	-0.05225		
BBB corporate OAS	9.556e-05	0.00014	-2.105e-06		
High-yield corporate OAS	0.001522	0.002271	-2.263e-06		
WTI front-month future	6.762	20.98	-6.751		
Gold future	3901	4872	-316.8		
Broad trade-weighted dollar	0.05291	0.06738	-0.008505		
10y TIPS real yield	0.002027	0.0008289	-0.001457		
Brent front-month future	6.383	19.52	-5.434		
DXY dollar index	0.1103	0.06787	-0.05857		
EUR/USD	0.1107	0.1152	-0.004557		
USD/JPY	0.382	0.2325	-0.004279		
USD/CHF	0.195	0.1764	-0.0006492		
VIX	2.082	1.577	-0.3673		
20y+ Treasury ETF (TLT)	0.436	0.3004	-0.05973		
HY bond ETF (HYG)	0.06618	0.1025	-0.02887		
Euro Stoxx 50	0.5974	1.59	-0.2746		
Nikkei 225	3.337	2.955	-0.07716		
EM equities (EEM)	3.312	3.787	-0.03518		
Tel Aviv 35	1.157	1.626	-0.004299		
Saudi equities (KSA ETF)	0.863	1.088	-0.01149		
US defense ETF (ITA)	2.18	2.914	-0.01881		
US airlines ETF (JETS)	3.059	6.953	-0.7265		
US energy ETF (XLE)	1.296	2.925	-0.113		
Henry Hub natgas future	5.884	7.728	-0.195		
Dry-bulk shipping ETF (BDRY)	3.615	6.873	-0.3091		
Bitcoin	4.821	6.185	-0.1107		

Table 2 (S4_prewar). Estimated impact of an increase in war risk, normalised to a 25 bp drop in the two-year Treasury yield

S4 pre-war window (2 Jan - 27 Feb): top-36% NLP index, as in the paper's 17/47 design. Window 2026-01-02 to 2026-02-27; nH = 14, nL = 14. Two-year yield variance: L days 0.00110, H days 0.00174. Absolute t-statistics (heteroskedasticity-robust) in parentheses.

Variable	Units	IV with nu1	IV with nu2	IV with nu3
Ten-year Treasury yield	pp chg	-0.241 (1.51)	0.0576 (0.14)	-0.0844 (0.39)
10y break-even inflation	pp chg	-0.124 (0.91)	-0.0845 (0.82)	-0.102 (1.39)
S&P 500	pct chg	-9.19 (0.83)	0.681 (0.12)	-0.482 (0.10)
BBB corporate OAS	pp chg	0.12 (0.79)	0.148 (2.59)	0.143 (2.14)
High-yield corporate OAS	pp chg	0.726 (0.93)	0.431 (3.11)	0.369 (2.50)
WTI front-month future	\$ chg	-0.684 (0.07)	-92.2 (0.07)	-0.449 (0.05)
Gold future	\$ chg	761 (0.53)	-4.38e+03 (0.95)	-1.14e+03 (0.64)
Broad trade-weighted dollar	pct chg	-1.09 (0.46)	-2.59 (0.58)	-1.43 (0.67)
10y TIPS real yield	pp chg	-0.116 (1.34)	0.431 (0.39)	-0.146 (1.50)
Brent front-month future	\$ chg	-0.0747 (0.01)	-1.77e+03 (0.01)	0.591 (0.05)
DXY dollar index	pct chg	-1.88 (0.57)	2.35 (0.31)	-0.765 (0.32)
EUR/USD	pct chg	5.63 (0.65)	0.0791 (0.03)	1.12 (0.51)
USD/JPY	pct chg	-6.5 (0.54)	-7.09 (1.12)	-6.9 (1.81)
USD/CHF	pct chg	-6.34 (0.64)	1.93 (0.35)	-0.548 (0.14)
VIX	pt chg	18.5 (0.87)	3.83 (0.45)	3.55 (0.41)
20y+ Treasury ETF (TLT)	pct chg	2.93 (1.41)	-6.39 (0.41)	1.42 (0.50)
HY bond ETF (HYG)	pct chg	-0.982 (0.54)	-1.73 (0.57)	-1.32 (1.12)
Euro Stoxx 50	pct chg	-10.1 (0.80)	-1.88 (0.81)	-1.82 (0.78)
Nikkei 225	pct chg	1.1 (0.21)	43.4 (0.21)	2.12 (0.39)
EM equities (EEM)	pct chg	-4.57 (0.81)	12 (0.41)	-2.27 (0.43)
Tel Aviv 35	pct chg	-10.1 (0.77)	-3.42 (0.64)	-5.07 (1.34)
Saudi equities (KSA ETF)	pct chg	-5.05 (0.66)	-19.3 (0.97)	-10.1 (0.99)
US defense ETF (ITA)	pct chg	-19.1 (0.64)	3.2 (0.52)	-0.62 (0.12)
US airlines ETF (JETS)	pct chg	-34.3 (0.66)	-13.3 (1.43)	-15 (1.86)
US energy ETF (XLE)	pct chg	7.23 (0.42)	-4.4 (0.35)	1.96 (0.34)
Henry Hub natgas future	pct chg	152 (0.66)	-189 (0.92)	-76.7 (0.50)
Dry-bulk shipping ETF (BDRY)	pct chg	8.14 (0.42)	-45.2 (0.42)	0.787 (0.05)
Bitcoin	pct chg	-82.5 (0.77)	-34.2 (2.95)	-33 (2.66)

Table 3 (S4_prewar). Variances and share explained by the war-risk factor

Variable	Var. on L days	Var. on H days	Predicted change in var.	% explained, H days	% explained, all days
Ten-year Treasury yield	0.001321	0.001371	7.246e-05	5.283	2.833
10y break-even inflation	0.0002571	0.0003714	0.0001067	28.71	17.93
S&P 500	0.7419	0.6638	0.002363	0.3561	0.2037
BBB corporate OAS	9.286e-05	0.0002786	0.0002077	74.55	62.95
High-yield corporate OAS	0.001843	0.004329	0.001383	31.94	26.86
WTI front-month future	1.649	1.952	0.002047	0.1049	0.05346
Gold future	3.865e+04	7728	1.331e+04	172.2	35.96
Broad trade-weighted dollar	0.06852	0.08994	0.0209	23.24	16.23
10y TIPS real yield	0.0009929	0.0006143	0.0002167	35.28	14.06

Variable	Var. on L days	Var. on H days	Predicted change in var.	% explained, H days	% explained, all days
Brent front-month future	1.817	2.7	0.003557	0.1318	0.07871
DXY dollar index	0.2011	0.1473	0.005955	4.044	2.068
EUR/USD	0.2281	0.2272	0.01281	5.639	3.387
USD/JPY	0.2945	0.7054	0.4846	68.7	57.4
USD/CHF	0.5134	0.3834	0.003052	0.7959	0.4114
VIX	2.113	2.764	0.1282	4.639	3.269
20y+ Treasury ETF (TLT)	0.4153	0.2902	0.02061	7.1	3.175
HY bond ETF (HYG)	0.01672	0.03235	0.01785	55.18	42.57
Euro Stoxx 50	0.3801	0.6757	0.03356	4.967	2.537
Nikkei 225	1.756	2.582	0.06236	2.416	0.9384
EM equities (EEM)	1.622	1.14	0.05246	4.6	1.867
Tel Aviv 35	0.7174	1.113	0.261	23.46	11.36
Saudi equities (KSA ETF)	0.4958	1.469	1.037	70.57	35.12
US defense ETF (ITA)	2.002	1.433	0.003912	0.2731	0.09096
US airlines ETF (JETS)	3.282	7.47	2.296	30.73	24.88
US energy ETF (XLE)	2.508	1.314	0.03914	2.98	0.7899
Henry Hub natgas future	385.9	120.7	59.87	49.62	15.05
Dry-bulk shipping ETF (BDRY)	8.431	4.914	0.006293	0.1281	0.04096
Bitcoin	7.523	32.5	11.08	34.1	34.09

Table 2 (S5_warpost). Estimated impact of an increase in war risk, normalised to a 25 bp drop in the two-year Treasury yield

S5 war and post-war window (2 Mar - 16 Sep). Window 2026-03-02 to 2026-09-16; nH = 35, nL = 35. Two-year yield variance: L days 0.00343, H days 0.00212. Absolute t-statistics (heteroskedasticity-robust) in parentheses.

Variable	Units	IV with nu1	IV with nu2	IV with nu3
Ten-year Treasury yield	pp chg	-0.0636 (0.32)	-0.676 (0.36)	0.0903 (0.35)
10y break-even inflation	pp chg	0.302 (0.59)	-0.0742 (1.20)	-0.00303 (0.06)
S&P 500	pct chg	0.867 (0.22)	-2.91 (0.06)	1.21 (0.40)
BBB corporate OAS	pp chg	0.0635 (0.73)	0.0189 (0.31)	0.0333 (0.83)
High-yield corporate OAS	pp chg	0.488 (0.64)	-0.0799 (0.38)	0.0124 (0.07)
WTI front-month future	\$ chg	48.8 (0.54)	-43.6 (2.21)	-22.6 (1.37)
Gold future	\$ chg	382 (0.70)	-16.6 (0.03)	135 (0.43)
Broad trade-weighted dollar	pct chg	-1.41 (0.80)	1.2 (0.33)	0.0876 (0.04)
10y TIPS real yield	pp chg	-0.365 (0.97)	-0.325 (2.79)	-0.312 (3.55)
Brent front-month future	\$ chg	46.9 (0.54)	-42.6 (2.92)	-21.7 (1.73)
DXY dollar index	pct chg	-2.94 (0.84)	-1.84 (2.07)	-1.68 (1.77)
EUR/USD	pct chg	-0.397 (0.30)	4.85 (0.21)	-0.261 (0.20)
USD/JPY	pct chg	-0.755 (0.33)	-25.9 (0.36)	-2.59 (0.58)
USD/CHF	pct chg	0.372 (0.23)	4.07 (0.10)	0.394 (0.25)
VIX	pt chg	-4.11 (0.50)	-13.6 (0.65)	-5.25 (0.57)
20y+ Treasury ETF (TLT)	pct chg	-2.19 (0.36)	-5.38 (0.34)	-3.5 (0.90)
HY bond ETF (HYG)	pct chg	-1.16 (0.32)	4.25 (1.05)	1.12 (1.35)
Euro Stoxx 50	pct chg	-11.8 (0.53)	11.3 (1.80)	5.75 (1.08)
Nikkei 225	pct chg	-2.77 (0.45)	-31.9 (0.46)	-4.18 (0.58)
EM equities (EEM)	pct chg	-4.7 (0.27)	9.34 (0.42)	-0.638 (0.11)

Variable	Units	IV with nu1	IV with nu2	IV with nu3
Tel Aviv 35	pct chg	4.83 (0.40)	-11 (0.29)	-0.899 (0.07)
Saudi equities (KSA ETF)	pct chg	-3.7 (0.40)	21.7 (0.96)	6.27 (0.86)
US defense ETF (ITA)	pct chg	-13.8 (0.50)	11.6 (1.72)	2.89 (0.59)
US airlines ETF (JETS)	pct chg	-30.9 (0.52)	17.4 (3.72)	6.66 (1.64)
US energy ETF (XLE)	pct chg	23.3 (0.55)	-8.36 (2.25)	-2.7 (0.81)
Henry Hub natgas future	pct chg	36.7 (0.61)	-6.19 (0.71)	-1.19 (0.15)
Dry-bulk shipping ETF (BDRY)	pct chg	11.3 (0.47)	-36.2 (0.90)	-11 (0.62)
Bitcoin	pct chg	14.2 (0.75)	-7.82 (0.33)	0.338 (0.02)

Table 3 (S5_warpost). Variances and share explained by the war-risk factor

Variable	Var. on L days	Var. on H days	Predicted change in var.	% explained, H days	% explained, all days
Ten-year Treasury yield	0.002357	0.001709	-0.0001706		
10y break-even inflation	0.0003743	0.0005886	-1.92e-07		
S&P 500	0.6864	0.8328	-0.03123		
BBB corporate OAS	9.714e-05	9.429e-05	-2.319e-05		
High-yield corporate OAS	0.001709	0.002143	-3.194e-06		
WTI front-month future	5.989	25.91	-10.89		
Gold future	4372	4430	-388.1		
Broad trade-weighted dollar	0.05849	0.07958	-0.0001425		
10y TIPS real yield	0.002137	0.0007486	-0.002043		
Brent front-month future	5.547	24.19	-10.11		
DXY dollar index	0.1234	0.06836	-0.06059		
EUR/USD	0.1046	0.1213	-0.001426		
USD/JPY	0.4538	0.2531	-0.1407		
USD/CHF	0.223	0.2043	-0.003245		
VIX	2.161	1.581	-0.5881		
20y+ Treasury ETF (TLT)	0.4047	0.2788	-0.2613		
HY bond ETF (HYG)	0.07269	0.1127	-0.02683		
Euro Stoxx 50	0.6256	1.909	-0.7807		
Nikkei 225	4.052	3.239	-0.3489		
EM equities (EEM)	3.559	4.444	-0.008709		
Tel Aviv 35	1.382	1.69	-0.01622		
Saudi equities (KSA ETF)	0.6294	1.323	-0.8401		
US defense ETF (ITA)	2.279	3.594	-0.178		
US airlines ETF (JETS)	2.877	7.932	-0.947		
US energy ETF (XLE)	1.525	3.291	-0.156		
Henry Hub natgas future	5.765	8.209	-0.03009		
Dry-bulk shipping ETF (BDRY)	4.034	7.765	-2.59		
Bitcoin	5.105	6.158	-0.002389		

Table 2 (S6_curated). Estimated impact of an increase in war risk, normalised to a 25 bp drop in the two-year Treasury yield

S6 hand-curated event days (the paper's 'reading newspapers' approach). Window 2026-01-02 to 2026-09-17; nH = 49, nL = 49. Two-year yield variance: L days 0.00296, H days 0.00266. Absolute

t-statistics (heteroskedasticity-robust) in parentheses.

Variable	Units	IV with nu1	IV with nu2	IV with nu3
Ten-year Treasury yield	pp chg	-0.0889 (0.26)	0.27 (0.08)	-0.211 (0.65)
10y break-even inflation	pp chg	0.161 (0.27)	-0.214 (0.70)	-0.0798 (0.40)
S&P 500	pct chg	5.35 (0.39)	-7.27 (0.35)	-3.11 (0.22)
BBB corporate OAS	pp chg	-0.0493 (0.24)	0.291 (0.45)	0.0971 (0.40)
High-yield corporate OAS	pp chg	-0.229 (0.24)	1.61 (0.52)	0.632 (0.53)
WTI front-month future	\$ chg	34.3 (0.28)	-77.9 (1.21)	-44.6 (1.00)
Gold future	\$ chg	934 (0.35)	-1.11e+03 (0.70)	-768 (0.54)
Broad trade-weighted dollar	pct chg	-2.1 (0.14)	2.53 (0.16)	1.35 (0.11)
10y TIPS real yield	pp chg	-0.25 (0.65)	-0.254 (1.17)	-0.255 (1.20)
Brent front-month future	\$ chg	27.2 (0.26)	-90 (0.96)	-45.6 (0.88)
DXY dollar index	pct chg	0.37 (0.09)	-20.5 (0.12)	-2.24 (0.47)
EUR/USD	pct chg	-3 (0.33)	0.398 (0.10)	-0.382 (0.13)
USD/JPY	pct chg	3.98 (0.32)	8.89 (0.84)	7.48 (0.90)
USD/CHF	pct chg	3.03 (0.35)	5.42 (0.65)	4.71 (0.72)
VIX	pt chg	-16.4 (0.37)	6.17 (0.31)	3.05 (0.17)
20y+ Treasury ETF (TLT)	pct chg	1.14 (0.23)	-24.3 (0.15)	3.71 (0.38)
HY bond ETF (HYG)	pct chg	0.33 (0.13)	-16.8 (0.10)	2.12 (0.45)
Euro Stoxx 50	pct chg	-2.81 (0.21)	27.1 (0.35)	4.52 (0.30)
Nikkei 225	pct chg	69.4 (0.05)	12.7 (0.37)	14.3 (0.43)
EM equities (EEM)	pct chg	2.57 (0.13)	118 (0.13)	1.35 (0.07)
Tel Aviv 35	pct chg	4.87 (0.25)	-38.9 (0.39)	-14.2 (0.32)
Saudi equities (KSA ETF)	pct chg	5.95 (0.35)	-16.3 (0.49)	-8.15 (0.37)
US defense ETF (ITA)	pct chg	21 (0.34)	-6.23 (0.57)	-5 (0.47)
US airlines ETF (JETS)	pct chg	26.9 (0.37)	-7.96 (0.48)	-7.81 (0.47)
US energy ETF (XLE)	pct chg	12.5 (0.31)	-9.51 (0.81)	-3.55 (0.38)
Henry Hub natgas future	pct chg	-80 (0.33)	-160 (1.07)	-147 (1.01)
Dry-bulk shipping ETF (BDRY)	pct chg	0.584 (0.03)	-247 (0.03)	0.426 (0.02)
Bitcoin	pct chg	26.8 (0.32)	-13.8 (0.45)	-5.26 (0.20)

Table 3 (S6_curated). Variances and share explained by the war-risk factor

Variable	Var. on L days	Var. on H days	Predicted change in var.	% explained, H days	% explained, all days
Ten-year Treasury yield	0.002	0.002149	-0.0002138		
10y break-even inflation	0.0004184	0.0005898	-3.055e-05		
S&P 500	0.6788	0.8689	-0.04647		
BBB corporate OAS	0.0001122	0.0001837	-4.53e-05		
High-yield corporate OAS	0.001751	0.003588	-0.001916		
WTI front-month future	9.13	22.62	-9.532		
Gold future	6951	1.274e+04	-2828		
Broad trade-weighted dollar	0.0846	0.09528	-0.003535		
10y TIPS real yield	0.001537	0.001233	-0.0003116		
Brent front-month future	9.978	22.4	-9.979		
DXY dollar index	0.09706	0.1382	-0.02417		
EUR/USD	0.124	0.1374	-0.0007019		
USD/JPY	0.384	0.2094	-0.2687		
USD/CHF	0.2293	0.165	-0.1064		
VIX	2.751	3.286	-0.04477		
20y+ Treasury ETF (TLT)	0.3403	0.4865	-0.0661		
HY bond ETF (HYG)	0.06662	0.09475	-0.02165		

Variable	Var. on L days	Var. on H days	Predicted change in var.	% explained, H days	% explained, all days
Euro Stoxx 50	1.117	1.499	-0.09813		
Nikkei 225	4.722	4.003	-0.05971		
EM equities (EEM)	4.654	3.086	-0.008691		
Tel Aviv 35	0.9761	1.749	-0.9795		
Saudi equities (KSA ETF)	0.7575	1.236	-0.3186		
US defense ETF (ITA)	2.048	2.685	-0.1201		
US airlines ETF (JETS)	4.437	5.51	-0.2929		
US energy ETF (XLE)	1.776	2.342	-0.06062		
Henry Hub natgas future	93.51	26.71	-103.9		
Dry-bulk shipping ETF (BDRY)	6.868	7.205	-0.0008696		
Bitcoin	8.017	9.854	-0.133		

Table 2 (S3_surprise_q75). Estimated impact of an increase in war risk, normalised to a 25 bp drop in the two-year Treasury yield

S3 surprise index (intensity relative to the trailing 10-day median). Window 2026-01-02 to 2026-09-17; nH = 45, nL = 45. Two-year yield variance: L days 0.00293, H days 0.00218. Absolute t-statistics (heteroskedasticity-robust) in parentheses.

Variable	Units	IV with nu1	IV with nu2	IV with nu3
Ten-year Treasury yield	pp chg	-0.278 (1.30)	-0.374 (2.10)	-0.374 (2.10)
10y break-even inflation	pp chg	0.0846 (0.37)	-0.128 (0.62)	-0.00204 (0.02)
S&P 500	pct chg	4.59 (0.60)	2.72 (0.41)	3.48 (1.08)
BBB corporate OAS	pp chg	0.0305 (0.50)	-0.0335 (0.12)	0.0248 (0.42)
High-yield corporate OAS	pp chg	-0.0399 (0.13)	5.58 (0.15)	0.339 (0.61)
WTI front-month future	\$ chg	1.56 (0.07)	-472 (0.08)	-23.1 (0.70)
Gold future	\$ chg	597 (0.67)	-1.55e+03 (0.70)	-660 (0.48)
Broad trade-weighted dollar	pct chg	-1.09 (0.77)	1.56 (0.46)	-0.0795 (0.06)
10y TIPS real yield	pp chg	-0.362 (0.97)	-0.249 (2.53)	-0.222 (1.88)
Brent front-month future	\$ chg	2.24 (0.10)	-433 (0.11)	-27.5 (0.73)
DXY dollar index	pct chg	-1.12 (0.71)	2.77 (0.34)	-0.485 (0.30)
EUR/USD	pct chg	-0.836 (0.40)	-1.03 (0.12)	-0.859 (0.43)
USD/JPY	pct chg	0.363 (0.14)	-4.18 (0.06)	0.307 (0.12)
USD/CHF	pct chg	-0.16 (0.07)	4.71 (0.04)	-0.118 (0.05)
VIX	pt chg	-10.5 (0.56)	0.95 (0.10)	-2.18 (0.34)
20y+ Treasury ETF (TLT)	pct chg	2.07 (0.75)	7.2 (0.85)	1.61 (0.59)
HY bond ETF (HYG)	pct chg	0.555 (0.37)	-5 (0.20)	1.11 (0.69)
Euro Stoxx 50	pct chg	-2.68 (0.32)	16 (0.53)	2.55 (0.40)
Nikkei 225	pct chg	3.11 (0.35)	56.9 (0.42)	10.8 (0.71)
EM equities (EEM)	pct chg	0.994 (0.10)	60.4 (0.10)	-1.41 (0.16)
Tel Aviv 35	pct chg	20.3 (0.46)	2.51 (0.45)	4.19 (0.87)
Saudi equities (KSA ETF)	pct chg	-2.24 (0.33)	12.4 (0.42)	0.627 (0.12)
US defense ETF (ITA)	pct chg	5.68 (0.61)	-5.54 (0.21)	2.87 (0.35)
US airlines ETF (JETS)	pct chg	3.22 (0.27)	-120 (0.22)	6.4 (0.47)
US energy ETF (XLE)	pct chg	8.89 (0.44)	-7.51 (0.73)	-0.356 (0.06)
Henry Hub natgas future	pct chg	-59.7 (0.56)	-114 (0.70)	-95.8 (0.77)
Dry-bulk shipping ETF (BDRY)	pct chg	3.4 (0.24)	-79.5 (0.28)	-2.54 (0.19)
Bitcoin	pct chg	41.3 (0.60)	-29.9 (2.26)	-20.9 (1.49)

Table 3 (S3_surprise_q75). Variances and share explained by the war-risk factor

Variable	Var. on L days	Var. on H days	Predicted change in var.	% explained, H days	% explained, all days
Ten-year Treasury yield	0.00252	0.001667	-0.001673		
10y break-even inflation	0.0004267	0.0005156	-4.979e-08		
S&P 500	0.6626	0.6379	-0.147		
BBB corporate OAS	0.0001267	0.0001378	-7.359e-06		
High-yield corporate OAS	0.001249	0.002991	-0.001371		
WTI front-month future	6.496	11.86	-6.44		
Gold future	6826	1.354e+04	-5270		
Broad trade-weighted dollar	0.05309	0.0715	-6.885e-05		
10y TIPS real yield	0.001791	0.001004	-0.0005888		
Brent front-month future	7.255	14.35	-9.152		
DXY dollar index	0.1002	0.1181	-0.00285		
EUR/USD	0.1472	0.1365	-0.008821		
USD/JPY	0.3458	0.3605	-0.001125		
USD/CHF	0.2349	0.2399	-0.000165		
VIX	1.649	1.688	-0.05778		
20y+ Treasury ETF (TLT)	0.4512	0.33	-0.03156		
HY bond ETF (HYG)	0.07276	0.08812	-0.01497		
Euro Stoxx 50	0.7335	1.079	-0.08059		
Nikkei 225	3.491	2.017	-1.265		
EM equities (EEM)	3.007	2.861	-0.02399		
Tel Aviv 35	1.826	1.439	-0.1821		
Saudi equities (KSA ETF)	0.9602	1.125	-0.00476		
US defense ETF (ITA)	2.374	2.563	-0.0994		
US airlines ETF (JETS)	3.761	6.694	-0.4957		
US energy ETF (XLE)	2.028	2.366	-0.001533		
Henry Hub natgas future	102.6	45.78	-111.1		
Dry-bulk shipping ETF (BDRY)	5.167	6.828	-0.07787		
Bitcoin	4.388	13.73	-5.243		

Table 2 (S7_volume_q75). Estimated impact of an increase in war risk, normalised to a 25 bp drop in the two-year Treasury yield

S7 volume-only index (headline count + Wikipedia text volume), which concentrates on the sustained-war weeks. Window 2026-01-02 to 2026-09-17; nH = 45, nL = 45. Two-year yield variance: L days 0.00322, H days 0.00201. Absolute t-statistics (heteroskedasticity-robust) in parentheses.

Variable	Units	IV with nu1	IV with nu2	IV with nu3
Ten-year Treasury yield	pp chg	-0.246 (1.97)	-0.281 (3.28)	-0.277 (3.35)
10y break-even inflation	pp chg	-0.0181 (0.23)	-0.24 (0.21)	-0.00849 (0.10)
S&P 500	pct chg	5.66 (1.18)	3.56 (0.96)	4.19 (1.59)
BBB corporate OAS	pp chg	-0.0656 (0.82)	0.0489 (0.63)	-0.00299 (0.07)
High-yield corporate OAS	pp chg	-0.175 (0.71)	0.546 (0.71)	0.0446 (0.19)
WTI front-month future	\$ chg	-9.25 (0.81)	81.9 (0.50)	-3.65 (0.28)

Variable	Units	IV with nu1	IV with nu2	IV with nu3
Gold future	\$ chg	163 (0.74)	88.6 (0.11)	162 (0.73)
Broad trade-weighted dollar	pct chg	-1.06 (1.21)	1.02 (0.34)	-0.502 (0.57)
10y TIPS real yield	pp chg	-0.228 (1.82)	-0.124 (1.77)	-0.122 (1.70)
Brent front-month future	\$ chg	-11.3 (0.86)	76.9 (0.61)	-1.66 (0.11)
DXY dollar index	pct chg	-2.4 (1.32)	-2.2 (2.30)	-2.23 (2.44)
EUR/USD	pct chg	1.55 (0.87)	-3.49 (0.97)	-0.662 (0.42)
USD/JPY	pct chg	-2.75 (0.90)	3.16 (0.66)	0.208 (0.09)
USD/CHF	pct chg	-2.62 (0.85)	1.71 (0.52)	-0.289 (0.16)
VIX	pt chg	-10.2 (1.02)	-7.25 (0.90)	-8.23 (1.46)
20y+ Treasury ETF (TLT)	pct chg	1.22 (0.74)	7.94 (0.64)	0.795 (0.44)
HY bond ETF (HYG)	pct chg	1.18 (1.49)	1.07 (0.65)	1.15 (1.40)
Euro Stoxx 50	pct chg	-0.663 (0.19)	95.9 (0.22)	2.88 (0.67)
Nikkei 225	pct chg	0.334 (0.09)	-374 (0.09)	-0.0912 (0.02)
EM equities (EEM)	pct chg	9.19 (1.26)	12.8 (1.39)	10.7 (1.70)
Tel Aviv 35	pct chg	3.92 (0.81)	-2.99 (0.24)	2.11 (0.57)
Saudi equities (KSA ETF)	pct chg	2.07 (0.67)	-15.2 (0.48)	0.35 (0.10)
US defense ETF (ITA)	pct chg	4.29 (0.84)	7.51 (0.51)	4.54 (0.89)
US airlines ETF (JETS)	pct chg	7.99 (1.03)	-6.52 (0.30)	4.41 (0.62)
US energy ETF (XLE)	pct chg	4.27 (0.51)	-29 (0.90)	-5.16 (0.80)
Henry Hub natgas future	pct chg	1.66 (0.21)	-24.4 (0.20)	0.896 (0.13)
Dry-bulk shipping ETF (BDRY)	pct chg	0.357 (0.05)	-148 (0.05)	0.206 (0.03)
Bitcoin	pct chg	9.21 (1.00)	-4.79 (0.21)	4.89 (0.64)

Table 3 (S7_volume_q75). Variances and share explained by the war-risk factor

Variable	Var. on L days	Var. on H days	Predicted change in var.	% explained, H days	% explained, all days
Ten-year Treasury yield	0.002298	0.001262	-0.001483		
10y break-even inflation	0.0005133	0.00046	-1.394e-06		
S&P 500	0.9489	0.6908	-0.344		
BBB corporate OAS	0.0001178	0.00016	-1.724e-07		
High-yield corporate OAS	0.002213	0.00336	-3.845e-05		
WTI front-month future	9.824	19.5	-0.261		
Gold future	5217	4940	-514.3		
Broad trade-weighted dollar	0.0744	0.09604	-0.004685		
10y TIPS real yield	0.001384	0.0009	-0.0002865		
Brent front-month future	12.46	23.04	-0.05425		
DXY dollar index	0.1391	0.06927	-0.09734		
EUR/USD	0.08387	0.1538	-0.008472		
USD/JPY	0.2335	0.355	-0.0008375		
USD/CHF	0.1946	0.254	-0.001615		
VIX	3.009	2.028	-1.327		
20y+ Treasury ETF (TLT)	0.391	0.2519	-0.01238		
HY bond ETF (HYG)	0.08806	0.0701	-0.02612		
Euro Stoxx 50	0.8786	1.666	-0.173		
Nikkei 225	3.05	5.402	-0.0001735		
EM equities (EEM)	4.706	3.261	-2.234		
Tel Aviv 35	1.456	1.666	-0.0908		
Saudi equities (KSA ETF)	0.6769	1.058	-0.002402		
US defense ETF (ITA)	2.765	2.337	-0.404		
US airlines ETF (JETS)	4.127	4.69	-0.3806		
US energy ETF (XLE)	1.706	3.23	-0.5222		

Variable	Var. on L days	Var. on H days	Predicted change in var.	% explained, H days	% explained, all days
Henry Hub natgas future	4.987	5.662	-0.01575		
Dry-bulk shipping ETF (BDRY)	4.932	5.324	-0.0008328		
Bitcoin	5.538	6.106	-0.4619		

Table 2b (S1, alternative normalisation). Impact of a war-risk increase that raises the Brent front-month future by \$5

Brent variance: L days 6.38, H days 19.52 ($\2).

Variable	Units	IV with nu1	IV with nu2	IV with nu3	% explained, H days	% explained, all days
Ten-year Treasury yield	pp chg	0.0178 (1.68)	-0.0783 (0.58)	0.0288 (2.97)	26.8	3.0
10y break-even inflation	pp chg	0.0164 (3.59)	0.0159 (1.18)	0.0164 (3.66)	27.3	4.1
S&P 500	pct chg	-0.609 (3.70)	0.208 (0.24)	-0.595 (3.66)	25.8	3.4
BBB corporate OAS	pp chg	0.00527 (1.59)	0.0124 (0.72)	0.00574 (1.67)	12.4	1.6
High-yield corporate OAS	pp chg	0.0359 (2.69)	0.0379 (1.23)	0.0361 (2.71)	30.2	3.4
WTI front-month future	\$ chg	5.06 (8.15)	5.37 (8.39)	5.14 (8.16)	66.2	13.6
Gold future	\$ chg	-8.45 (0.32)	-243 (0.30)	-4.42 (0.16)	0.2	0.0
Broad trade-weighted dollar	pct chg	0.22 (3.21)	0.0796 (0.38)	0.204 (2.84)	35.5	4.2
10y TIPS real yield	pp chg	0.00138 (0.14)	-1.38 (0.13)	0.0137 (1.43)	11.9	1.0
DXY dollar index	pct chg	0.184 (3.19)	-0.397 (0.83)	0.157 (2.68)	19.1	1.4
EUR/USD	pct chg	-0.169 (1.25)	-0.103 (0.19)	-0.164 (1.23)	12.3	1.3
USD/JPY	pct chg	-0.0423 (0.22)	6.17 (0.23)	-0.0135 (0.07)	0.0	0.0
USD/CHF	pct chg	0.0558 (0.38)	-0.325 (0.09)	0.0586 (0.40)	1.0	0.1
VIX	pt chg	0.904 (2.28)	-1.13 (0.44)	0.817 (2.05)	22.2	1.9
20y+ Treasury ETF (TLT)	pct chg	-0.179 (1.29)	1.27 (0.50)	-0.285 (2.17)	14.2	1.5
HY bond ETF (HYG)	pct chg	-0.192 (2.68)	-0.397 (1.54)	-0.192 (2.68)	18.9	3.4
Euro Stoxx 50	pct chg	-1.16 (4.09)	-1.41 (5.36)	-1.2 (4.52)	51.4	9.4
Nikkei 225	pct chg	-0.706 (0.75)	2.01 (0.35)	-0.596 (0.66)	5.5	0.4
EM equities (EEM)	pct chg	-1.35 (3.45)	-0.116 (0.06)	-1.29 (3.36)	23.1	3.5
Tel Aviv 35	pct chg	0.188 (0.43)	9.77 (0.37)	0.431 (1.03)	4.0	0.7
Saudi equities (KSA ETF)	pct chg	-0.466 (1.61)	-1.16 (0.90)	-0.45 (1.54)	9.8	1.3
US defense ETF (ITA)	pct chg	-1.11 (2.73)	-1.32 (1.11)	-1.11 (2.74)	22.1	3.4
US airlines ETF (JETS)	pct chg	-2.48 (4.74)	-2.92 (3.53)	-2.53 (4.68)	48.4	9.2
US energy ETF (XLE)	pct chg	1.47 (2.92)	2.14 (4.18)	1.53 (3.14)	41.8	7.2
Henry Hub natgas future	pct chg	1.41 (2.13)	1.88 (0.58)	1.44 (2.32)	14.1	0.3
Dry-bulk shipping ETF (BDRY)	pct chg	0.442 (0.73)	15 (0.76)	0.725 (1.11)	4.0	0.6
Bitcoin	pct chg	-0.409 (0.65)	-6.25 (0.46)	-0.4 (0.63)	1.4	0.1

Comparison with Rigobon and Sack (2003), Iraq, Jan 6 - Mar 17 2003 (combined-instrument estimates)

2003 and the 2026 pre-war window are normalised to a war-risk increase that lowers the two-year yield by 25 bp (the two-year yield variance rises on war days in both). The 2026 full-sample specifications are normalised to a war-risk increase that raises Brent by \$5, because the two-year yield variance does not rise on 2026 war days (see Section 3.2). |t| in parentheses.

Variable	Units	RS 2003 Iraq (-25bp 2y)	2026 pre-war S4 (-25bp 2y)	2026 S1 full sample (+\$5 Brent)	2026 S5 war/post (+\$5 Brent)	2026 S6 curated (+\$5 Brent)
Ten-year Treasury yield	pp chg	-0.26 (11.85)	-0.0844 (0.39)	0.0288 (2.97)	0.0304 (3.65)	0.0259 (2.00)
Break-even inflation (10y)	pp chg	-0.11 (3.45)	-0.102 (1.39)	0.0164 (3.66)	0.0171 (4.46)	0.0174 (3.06)
S&P 500	pct chg	-3.76 (2.90)	-0.482 (0.10)	-0.595 (3.66)	-0.609 (4.64)	-0.701 (2.94)
BBB yield spread	pp chg	0.05 (3.79)	0.143 (2.14)	0.00574 (1.67)	0.00381 (1.41)	0.00669 (1.61)
High-yield spread	pp chg	0.34 (5.40)	0.369 (2.50)	0.0361 (2.71)	0.0333 (2.87)	0.0464 (2.93)
Oil price (12-month future)	\$ chg	0.77 (2.44)	-0.449 (0.05)	5.14 (8.16)	5.09 (9.00)	5.36 (8.47)
Gold price	\$ chg	1.3 (0.26)	-1.14e+03 (0.64)	-4.42 (0.16)	-18 (0.74)	12.6 (0.33)
Dollar (broad index)	pct chg	-0.44 (2.22)	-1.43 (0.67)	0.204 (2.84)	0.216 (3.49)	0.112 (1.04)

Note: the 2003 oil variable is the 12-month WTI future; here it is the front-month WTI future. Corporate spreads here are ICE BofA option-adjusted spreads (FRED) rather than Merrill Lynch index spreads to the Treasury curve.

Summary across specifications: combined-instrument (nu3) estimates, war-risk increase lowering the 2y yield 25 bp; |t| in parentheses

variable	units	S1_level_q7 5	S2_level_q8 5	S3_surprise_q 75	S4_prewar	S5_warpost	S6_curated	S7_volume_q 75
Ten-year Treasury yield	pp chg	-0.0667 (0.88)	0.121 (0.24)	-0.374 (2.10)	-0.0844 (0.39)	0.0903 (0.35)	-0.211 (0.65)	-0.277 (3.35)
10y break-even inflation	pp chg	0.0177 (0.44)	-0.0258 (0.34)	-0.00204 (0.02)	-0.102 (1.39)	-0.00303 (0.06)	-0.0798 (0.40)	-0.00849 (0.10)
S&P 500	pct chg	1.5 (0.87)	-0.886 (0.09)	3.48 (1.08)	-0.482 (0.10)	1.21 (0.40)	-3.11 (0.22)	4.19 (1.59)
BBB corporate OAS	pp chg	0.00957 (0.28)	-0.025 (0.17)	0.0248 (0.42)	0.143 (2.14)	0.0333 (0.83)	0.0971 (0.40)	-0.00299 (0.07)
High-yield corporate OAS	pp chg	0.00993 (0.07)	-0.275 (0.85)	0.339 (0.61)	0.369 (2.50)	0.0124 (0.07)	0.632 (0.53)	0.0446 (0.19)
WTI front-month future	\$ chg	-17 (1.19)	-21 (1.75)	-23.1 (0.70)	-0.449 (0.05)	-22.6 (1.37)	-44.6 (1.00)	-3.65 (0.28)
Gold future	\$ chg	117 (0.70)	-145 (0.42)	-660 (0.48)	-1.14e+03 (0.64)	135 (0.43)	-768 (0.54)	162 (0.73)
Broad trade-weighted dollar	pct chg	-0.616 (1.23)	0.399 (0.31)	-0.0795 (0.06)	-1.43 (0.67)	0.0876 (0.04)	1.35 (0.11)	-0.502 (0.57)

Summary across specifications: combined-instrument (nu3) estimates, war-risk increase raising Brent \$5; |t| in parentheses

variable	units	S1_level_q7 5	S2_level_q8 5	S3_surprise_q 75	S4_prewar	S5_warpost	S6_curated	S7_volume_q 75
Ten-year Treasury yield	pp chg	0.0288 (2.97)	0.0346 (2.85)	0.0296 (2.04)	-0.0258 (0.45)	0.0304 (3.65)	0.0259 (2.00)	0.0127 (0.75)

variable	units	S1_level_q7 5	S2_level_q8 5	S3_surprise_q 75	S4_prewar	S5_warpost	S6_curated	S7_volume_q 75
10y break-even inflation	pp chg	0.0164 (3.66)	0.0168 (2.53)	0.0168 (1.81)	-0.0013 (0.03)	0.0171 (4.46)	0.0174 (3.06)	0.00105 (0.15)
S&P 500	pct chg	-0.595 (3.66)	-0.469 (2.37)	-0.153 (0.49)	0.433 (0.24)	-0.609 (4.64)	-0.701 (2.94)	-0.474 (1.39)
BBB corporate OAS	pp chg	0.00574 (1.67)	0.000238 (0.08)	-0.000579 (0.12)	-0.000627 (0.01)	0.00381 (1.41)	0.00669 (1.61)	0.0105 (2.40)
High-yield corporate OAS	pp chg	0.0361 (2.71)	0.02 (1.88)	0.0408 (1.54)	-0.0718 (0.37)	0.0333 (2.87)	0.0464 (2.93)	0.0511 (1.98)
WTI front-month future	\$ chg	5.14 (8.16)	4.16 (11.06)	4.09 (9.19)	3.23 (8.89)	5.09 (9.00)	5.36 (8.47)	3.09 (1.34)
Gold future	\$ chg	-4.42 (0.16)	17 (0.58)	113 (1.28)	367 (1.01)	-18 (0.74)	12.6 (0.33)	-44.3 (1.85)
Broad trade-weighted dollar	pct chg	0.204 (2.84)	0.143 (2.54)	0.00371 (0.03)	-0.231 (0.43)	0.216 (3.49)	0.112 (1.04)	0.22 (1.94)
10y TIPS real yield	pp chg	0.0137 (1.43)	0.0205 (1.44)	0.0109 (0.72)	0.00157 (0.02)	0.0142 (1.74)	0.00865 (0.82)	-6.31e-05 (0.01)
DXY dollar index	pct chg	0.157 (2.68)	0.189 (2.74)	0.0825 (0.55)	-0.152 (0.15)	0.154 (3.20)	0.014 (0.10)	0.0842 (0.74)
EUR/USD	pct chg	-0.164 (1.23)	-0.04 (0.34)	-0.00868 (0.06)	0.331 (0.31)	-0.15 (1.25)	-0.152 (0.96)	-0.356 (2.91)
USD/JPY	pct chg	-0.0135 (0.07)	-0.0062 (0.02)	-0.231 (0.82)	-3 (1.23)	0.0133 (0.08)	0.0542 (0.28)	0.0487 (0.24)
USD/CHF	pct chg	0.0586 (0.40)	0.0921 (0.66)	-0.0152 (0.08)	-0.375 (0.26)	0.0515 (0.38)	0.0956 (0.68)	-0.052 (0.22)
VIX	pt chg	0.817 (2.05)	0.341 (1.31)	-0.266 (0.46)	-1.48 (0.34)	0.85 (2.52)	0.768 (1.26)	0.64 (0.91)
20y+ Treasury ETF (TLT)	pct chg	-0.285 (2.17)	-0.347 (2.43)	-0.272 (1.32)	0.0349 (0.03)	-0.335 (2.95)	-0.368 (1.49)	-0.0393 (0.19)
HY bond ETF (HYG)	pct chg	-0.192 (2.68)	-0.203 (1.85)	-0.00466 (0.03)	0.439 (1.74)	-0.232 (3.61)	-0.165 (1.68)	-0.0657 (0.41)
Euro Stoxx 50	pct chg	-1.2 (4.52)	-0.815 (2.86)	-0.529 (1.54)	1.63 (0.86)	-1.18 (4.88)	-0.93 (2.51)	-1.25 (3.21)
Nikkei 225	pct chg	-0.596 (0.66)	0.108 (0.10)	0.306 (0.27)	3.83 (1.09)	-0.499 (0.62)	-0.681 (0.83)	-2.47 (2.22)
EM equities (EEM)	pct chg	-1.29 (3.36)	-1.27 (2.32)	-0.316 (0.49)	0.436 (0.23)	-1.35 (4.07)	-0.945 (1.61)	-0.971 (1.16)
Tel Aviv 35	pct chg	0.431 (1.03)	-0.0843 (0.21)	0.297 (0.57)	1.4 (0.45)	-0.157 (0.45)	-0.547 (0.77)	-0.0253 (0.01)
Saudi equities (KSA ETF)	pct chg	-0.45 (1.54)	-0.754 (1.99)	-0.544 (1.05)	1.2 (0.58)	-0.552 (2.27)	-0.314 (0.81)	-0.607 (1.25)
US defense ETF (ITA)	pct chg	-1.11 (2.74)	-0.688 (1.24)	-0.428 (0.69)	0.948 (0.39)	-1.15 (3.25)	-0.203 (0.34)	-1.08 (1.86)
US airlines ETF (JETS)	pct chg	-2.53 (4.68)	-2.55 (3.42)	-3.12 (2.95)	-9.4 (1.45)	-2.43 (5.88)	-1.93 (2.91)	-1.84 (2.24)
US energy ETF (XLE)	pct chg	1.53 (3.14)	1 (1.41)	1.38 (1.58)	2.59 (1.64)	1.29 (3.16)	0.957 (2.15)	1.89 (3.02)
Henry Hub natgas future	pct chg	1.44 (2.32)	1.97 (2.66)	0.434 (0.06)	-21.9 (0.37)	1.57 (2.94)	0.165 (0.06)	0.193 (0.22)
Dry-bulk shipping ETF (BDRY)	pct chg	0.725 (1.11)	1.22 (1.39)	0.976 (0.99)	2.84 (0.50)	0.569 (1.10)	-0.126 (0.16)	0.0615 (0.07)
Bitcoin	pct chg	-0.4 (0.63)	-0.225 (0.24)	3.67 (1.16)	22.5 (0.95)	-0.521 (0.92)	-0.369 (0.45)	-0.0206 (0.03)